

Wireshark Experiments

Experiment 1: Basic Packet Capture and Analysis

1. Capture HTTP traffic by visiting various websites
2. Analyze Ethernet, IP, TCP, and HTTP headers
3. Identify source/destination MAC addresses, IP addresses, and port numbers
4. Examine packet sizes and timing information
5. Use basic display filters (http, tcp, ip)

Experiment 2: TCP Three-Way Handshake Analysis

1. Capture TCP traffic during web browsing or file transfer
2. Filter for TCP SYN, SYN-ACK, and ACK packets
3. Analyze sequence and acknowledgment numbers
4. Study TCP flags and window sizes
5. Examine connection termination (FIN/RST packets)
6. Calculate round-trip time (RTT)

Experiment 3: DNS Query and Response Analysis

1. Capture DNS traffic using various query types (A, AAAA, MX, NS, PTR)
2. Analyze DNS query structure and response codes
3. Study DNS caching behavior
4. Investigate DNS over HTTPS (DoH) vs traditional DNS
5. Examine DNS tunneling attempts
6. Analyze DNS amplification attack patterns

Experiment 4: HTTP vs HTTPS Traffic Analysis

1. Capture both HTTP and HTTPS traffic simultaneously
2. Analyze plaintext HTTP headers and payloads
3. Examine TLS handshake process for HTTPS
4. Study certificate exchange and validation
5. Compare traffic patterns and sizes
6. Investigate HTTP/2 and HTTP/3 protocols if available

Experiment 5: Network Performance and Congestion Analysis

1. Capture traffic during high and low network usage periods
2. Analyze throughput, latency, and packet loss
3. Study TCP window scaling and congestion control algorithms
4. Identify network congestion indicators
5. Examine Quality of Service (QoS) markings
6. Generate I/O graphs and flow statistics

Experiment 6: Wireless Network Security Analysis

1. Capture 802.11 wireless frames (requires monitor mode)

2. Analyze beacon frames, probe requests/responses
3. Study WPA/WPA2/WPA3 handshake processes
4. Examine wireless encryption and authentication
5. Investigate deauthentication attacks
6. Analyze roaming behavior between access points

Experiment 7: Network Intrusion Detection and Analysis

1. Generate or capture suspicious network traffic
2. Analyze port scanning patterns (Nmap, etc.)
3. Identify brute force attacks on services
4. Study DDoS attack patterns and mitigation
5. Examine malware communication patterns
6. Investigate data exfiltration attempts
7. Analyze SQL injection and XSS attacks in HTTP traffic

Experiment 8: Email Protocol Analysis (SMTP, POP3, IMAP)

1. Capture email traffic using different protocols
2. Analyze SMTP conversation and commands
3. Study POP3 vs IMAP differences
4. Examine email headers and routing information
5. Investigate STARTTLS and email encryption
6. Analyze spam and phishing email characteristics

Experiment 9: IPv6 Transition and Dual-Stack Analysis

1. Capture dual-stack IPv4/IPv6 traffic
2. Analyze IPv6 header structure and addressing
3. Study ICMPv6 and neighbor discovery protocol
4. Examine IPv6 tunneling mechanisms (6to4, Teredo)
5. Investigate DHCPv6 vs SLAAC addressing
6. Analyze IPv6 security implications

Experiment 10: Advanced Protocol Analysis (MPLS, BGP, OSPF)

1. Capture MPLS-labeled packets in service provider networks
2. Analyze BGP routing updates and path selection
3. Study OSPF LSA propagation and convergence
4. Examine VLAN tagging and trunk protocols
5. Investigate spanning tree protocol operation
6. Analyze load balancing and redundancy mechanisms

Additional Experiments (M-Tech)

Experiment 11: IoT Device Communication Analysis

1. Capture traffic from various IoT devices
2. Analyze MQTT, CoAP, and other IoT protocols
3. Study device authentication and encryption methods
4. Examine firmware update processes
5. Investigate privacy implications of IoT communications
6. Analyze IoT botnet command and control patterns

Experiment 12: Network Forensics and Incident Response

1. Analyze provided incident packet captures
2. Reconstruct attack timeline and methodology
3. Identify compromised systems and data exfiltration
4. Correlate network evidence with system logs
5. Generate forensics report with chain of custody
6. Develop incident response recommendations

Experiment 13: Software-Defined Networking (SDN) Analysis

1. Capture OpenFlow protocol messages
2. Analyze controller-switch handshake process
3. Study flow table modifications and updates
4. Examine network topology discovery
5. Investigate SDN security vulnerabilities
6. Analyze network programmability features

Experiment 14: Cloud Network Traffic Analysis

1. Capture traffic to/from cloud services (AWS, Azure, GCP)
2. Analyze API calls and authentication mechanisms
3. Study content delivery network (CDN) behavior
4. Examine cloud storage synchronization patterns
5. Investigate multi-cloud networking
6. Analyze cost implications of traffic patterns